TOR AMPLITUDE OVER THE BASE RING FOR A COMPLEX OVER A FLAT ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Tor amplitude of a complex over a ring, defined through the vanishing of the derived tensor product against any module outside the interval. defined earlier in the same section, source lines 17040-17098. Supplies the property that is transported from one ring to the other.
- A projective resolution over the target ring is a flat resolution over the base ring when the target is flat as a module over the base. used at proof lines 17110-17114. Justifies computing the derived tensor product over the base with the same resolution.
- The comparison statement for tor amplitude under a ring map, of which this is a special case. the author cites Lemma lemma-push-tor-amplitude at proof line 17109; that statement is counted separately by this lane. Explains the position of the lemma in the section.

Source reference. Original lemma 066J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7938–7938. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0B66: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14059–14059. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $A \to B$ be a ring map. Assume that B is flat as an A-module. Let $K^\bullet$ be a complex of B-modules. Let $a, b \in \mathbf{Z}$. If $K^\bullet$ as a complex of B-modules has tor amplitude in $[a, b]$, then $K^\bullet$ as a complex of A-modules has tor amplitude in $[a, b]$.*

Proof. This is a special case of Lemma 0B66, but can also be seen directly as follows. We have $K^\bullet \otimes^{\mathbf{L}}_A M = K^\bullet \otimes^{\mathbf{L}}_B (M \otimes_A B)$ since any projective resolution of $K^\bullet$ as a complex of B-modules is a flat resolution of $K^\bullet$ as a complex of A-modules and can be used to compute $K^\bullet \otimes^{\mathbf{L}}_A M$. □